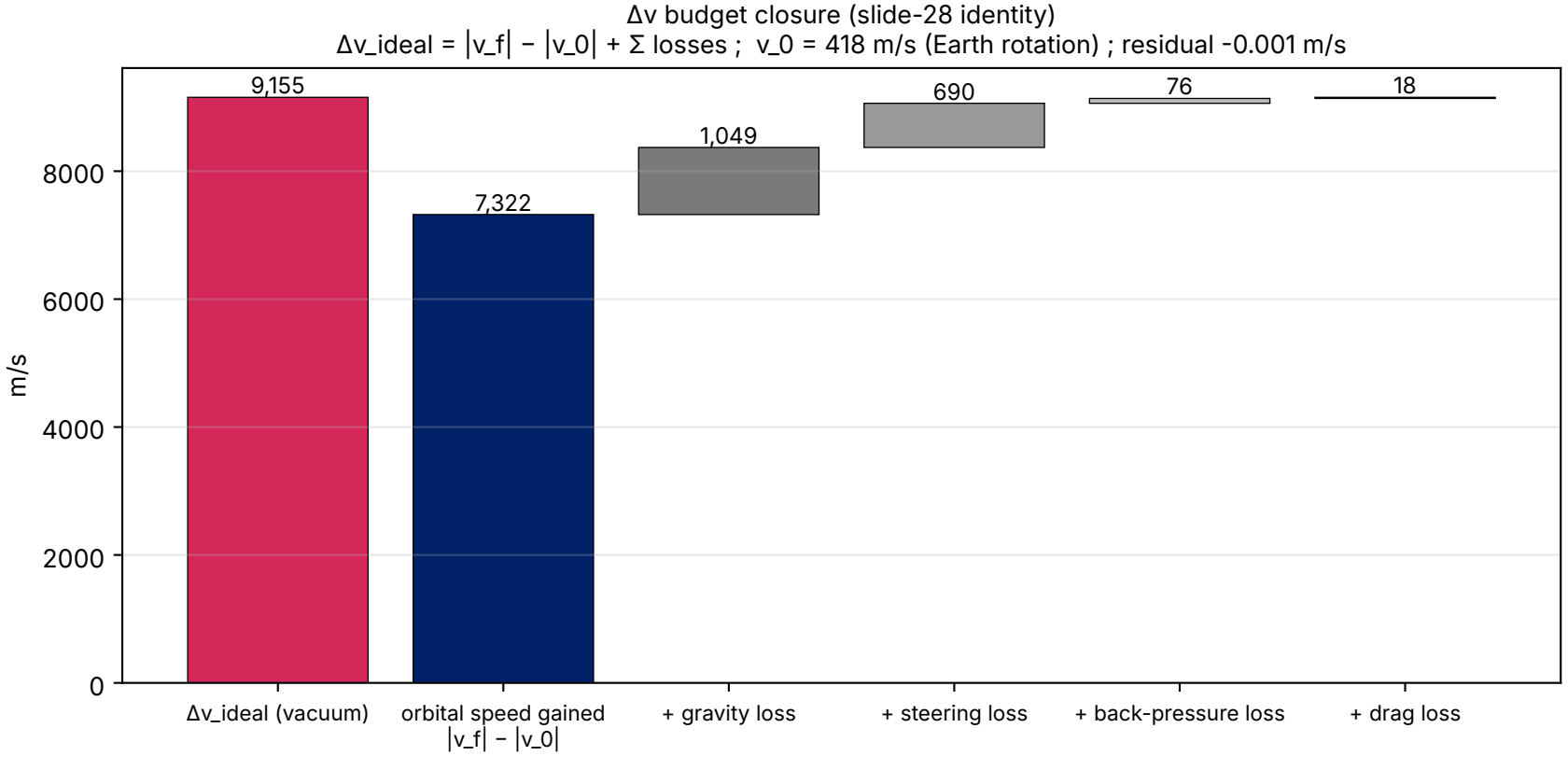
Δv budget closure (slide-28 identity)
Δv_ideal = |v_f| − |v_0| + Σ losses ; v_0 = 418 m/s (Earth rotation) ; residual -0.001 m/s
m/s
0
2000
4000
6000
8000
9,155
7,322
1,049
690
76
18
Δv_ideal (vacuum)
orbital speed gained
|v_f| − |v_0|
+ gravity loss
+ steering loss
+ back-pressure loss
+ drag loss